$\sqrt{2k}$

8 **A** θ is the □ the string makes with the vertical

$$T\cos\theta = mg$$

$$T\sin\theta = \frac{mv^2}{r}$$

$$= \frac{mv^2}{L\sin\theta}$$

$$\sin^2\theta = \frac{mv^2}{LT}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\frac{mv^2}{LT} + \left(\frac{mg}{T}\right)^2 = 1$$

$$\frac{(2.0)v^2}{(1.5)(30)} = 1 - \left[\frac{(2.0)(9.81)}{30}\right]^2$$

$$v = 3.5884$$

$$= 3.6 \text{ m s}^{-1}$$